

Operating Systems — Unit-wise & Topic-wise Repeated Question Bank

UNIT 1 — INTRODUCTION & PROCESS MANAGEMENT

Topic: Operating System Basics

1 Mark Questions

1. Operating System is a/an _____ between user and hardware of a system.
 2. The main purpose of an operating system is to:
 3. Which of the following is not a feature of Unix?
 4. Context-switch times are highly dependent on _____.
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Topic: PCB & Process

1 Mark Questions

1. A Program in execution is known as _____.
2. A process control block (PCB) also called a _____.
3. The Process Control Block contains all of the following except:

2 Mark Questions

1. What is the relationship between threads and processes?
2. Define context switching.
3. Differentiate between multitasking and multiprocessing operating systems.

6 Mark Questions

1. Explain the concept of process. Also describe the contents of a Process Control Block (PCB).
2. Explain the concept of process. Also describe the contents of a Process Control Block (PCB).

10 Mark Questions

1. Explain the functions of Operating System.
 2. Explain the features of Batch Operating System with advantages and disadvantages.
 3. Discuss the role of kernel in operating system. Explain the difference between monolithic kernel and microkernel.
 4. Illustrate the layered structure of an operating system and explain the role of each layer.
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Topic: Interrupts, System Calls & Virtual Machine

6 Mark Questions

1. What is System Call? Explain with example.
2. What is Time-sharing Operating System? How does it manage processes?
3. Explain the role of Operating System as I/O operation. Also explain the concept of context switching.

10 Mark Questions

1. Describe Interrupts and its different types. Explain Maskable and Non-maskable Interrupts.
 2. Describe the use of Fork and Executive System Calls. Explain Virtual Machine and its concept with history.
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UNIT 2 — PROCESS SCHEDULING

Topic: Scheduling Basics

1 Mark Questions

1. What type of scheduling is Round Robin scheduling?
2. Which module gives control of CPU to the process selected by short-term scheduler?
3. The primary distinction between short-term scheduler and long-term scheduler is _____.

4. Identify the primary design goal of Round Robin scheduling algorithm.
5. Identify the correct definition of context switch in process management.

2 Mark Questions

1. Define Swap-in and Swap-out.
2. Define Dispatch Latency.
3. Differentiate between preemptive and non-preemptive scheduling.

6 Mark Questions

1. Explain Shortest Job First CPU scheduling with example.
2. Explain Round Robin Scheduling with example.
3. Explain Round Robin Scheduling with example.
4. Explain FCFS and SJF scheduling algorithms with advantages and disadvantages.
5. Explain FCFS and SJF scheduling algorithms with advantages and disadvantages.

Topic: Scheduling Numericals

10 Mark Questions

1. Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of time listed. Use non-preemptive scheduling.

Process	Arrival Time	Burst Time
P1	0.0	8
P2	0.4	4
P3	1.0	1

- (a) Find average turnaround time using FCFS.
- (b) Find average turnaround time using SJF.
- (c) Compute average turnaround time if CPU is left idle for first 1 unit and then SJF scheduling is used.

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2. Explain the difference between preemptive and non-preemptive scheduling.

Four jobs A, B, C and D arrive at time 0.

Job	Burst Time
A	4
B	1
C	8
D	1

Find completion time of A under Round Robin scheduling with time slice = 1.

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3. Consider the set of 5 processes:

Process	Arrival Time	Burst Time
P1	03	01
P2	01	04
P3	04	02

P4	00	06
Process	Arrival	Burst
P5	Time	Time
	02	03

Using SJF non-preemptive scheduling calculate:

- Average Waiting Time
- Average Turnaround Time

Topic: Multilevel Scheduling

10 Mark Questions

- List and describe types of CPU schedulers with diagram.
- Distinguish between Multilevel Queue and Multilevel Feedback Queue Scheduling.

UNIT 3 – IPC, SYNCHRONIZATION & DEADLOCKS

Topic: IPC & Critical Section

1 Mark Questions

- Banker's Algorithm deals with _____.
- Critical Section must be followed by _____.
- For multiple instances of resource type which algorithm is used?
- Which one of the following is deadlock avoidance algorithm?
- A binary semaphore can have the following values.
- Strict alternation ensures synchronization using _____.

2 Mark Questions

- List Classical IPC problems.
- What is Inter-process communication?
- Explain how race conditions occur in concurrent systems.

6 Mark Questions

- How does mutex protect Critical Section from race condition?
- Why is deadlock state more critical than starvation?
- Describe Resource Allocation Graph with deadlock.
- Elaborate Readers-Writers problem in detail.
- Elaborate Readers-Writers problem in detail.

Topic: Synchronization Problems

10 Mark Questions

- Explain Dining-Philosophers Problem and give its solution.
- Explain Dining-Philosophers Problem and give its solution.
- What is Critical Section problem and what requirements should be satisfied?
- Give solution to 2-process Critical Section problem.
- Differentiate between Deadlock Avoidance and Deadlock Prevention.
- Explain Producer Consumer Problem.
- Explain methods used to solve process synchronization using Sleeping Barber problem.
- Explain Race condition with example. Also explain Dining Philosopher problem with figure.
- Explain Resource Allocation Graph. Discuss single instance and multi-instance RAG with suitable example.

UNIT 4 – MEMORY MANAGEMENT

Topic: Memory Management Basics

1 Mark Questions

1. CPU fetches instruction from memory according to value of _____.
2. Allocate the largest hole that is big enough is _____ memory allocation technique.
3. Which one of the following is address generated by CPU?
4. Memory management technique in which system stores and retrieves data from secondary storage for use in main memory is called?
5. Memory management is primarily handled by:
6. The anomaly that occurs in FIFO but not in LRU is:

2 Mark Questions

1. If size of Main Memory is 16 MB and block size is 4 KB. Find number of blocks and bits of physical address.
2. Define Compaction.
3. State purpose of Virtual Memory.

6 Mark Questions

1. Explain Segmentation with example.
2. Explain Segmentation with example.
3. Discuss process of overcoming Internal Fragmentation with example.
4. Explain when page faults occur and explain page fault handling procedure.

Topic: Page Replacement & Allocation Numericals

10 Mark Questions

1. Consider reference stream:

1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6

Find number of page faults using:

(a) FCFS (b) LRU

Using 2 frames.

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2. Explain difference between Internal and External Fragmentation with example.
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3. Explain Segmentation with Paging. What is Virtual Memory and give its advantages.
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4. How can we measure performance of Demand Paging?

Assume:

- Average page fault service time = 25 milliseconds
- Memory access time = 100 nanoseconds

Find Effective Access Time.

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5. Given reference string:

4, 7, 6, 1, 7, 6, 1, 2, 7, 2

Frame size = 3

Calculate number of page faults using:

(a) FIFO (b) Optimal (c) LRU

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6. Consider five memory partitions:

Partition Size
100 KB
500 KB
200 KB

Partition Size
300 KB
600 KB

Processes:

Process Size
212 KB
417 KB
112 KB
426 KB

Perform allocation using:

(a) First Fit (b) Best Fit (c) Worst Fit

UNIT 5 – I/O HARDWARE, FILE & DISK MANAGEMENT

Topic: UNIX & Linux Basics

1 Mark Questions

1. I/O hardware contains _____.
2. Which option is used with ls command to list directory attributes rather than contents?
3. Identify command used to remove empty directory.
4. What does chmod 777 filename do?

2 Mark Questions

1. Why SCAN algorithm is also known as Elevator Algorithm?
2. Explain all three file permissions in Linux.

6 Mark Questions

1. What is Disk Scheduling? Explain FCFS and SCAN disk scheduling algorithms.
2. Explain SSTF Disk Scheduling Algorithm with example.
3. Explain SSTF Disk Scheduling Algorithm with example.
4. Describe any six Linux commands with examples.

Topic: Disk Scheduling Numericals

10 Mark Questions

1. Explain different Disk Scheduling algorithms SCAN, C-SCAN and C-LOOK with example.

2. The available space list of computer memory is:

Starting Address	Block Size
100	50
200	150
450	600
1200	400

Requests:

25, 100, 250, 200, 100, 150

Use:

(a) First Fit (b) Best Fit (c) Worst Fit

3. Given disk queue requests:

98, 183, 41, 122, 14, 124, 65, 67

Initial head position = 53

Direction = Towards larger cylinder numbers

Cylinder range = 0 to 199

Calculate total head movement using:

(a) FCFS (b) SSTF (c) SCAN (d) C-SCAN (e) LOOK

Topic: UNIX & Shell Programming

10 Mark Questions

1. Explain features of UNIX in detail.
2. Explain features of UNIX in detail.
3. Write Shell Script which works as calculator and performs Addition, Subtraction, Division and Multiplication.
4. Given a file with permissions:

-rwxr-xr--

Explain each part and explain how to allow group write permission.